📖 Solutions Chapter Overview

A concise synthesis of three extensive video lectures that together cover the key concepts, calculations, and applications of solutions and dilute solutions in Class 12 Physical Chemistry, aimed at JEE preparation.

📚 Content Sources

- Section 1: Reviews **solution concentration**, **colligative properties**, and **Raoult's law** from a 223-minute YouTube video. (source)
- Section 2: Explores **ideal vs. non-ideal solutions**, **thermodynamic principles**, and **JEE-style problem solving** from a 393-minute YouTube video. (source)
- Section 3: Details **dilute solutions**, **osmotic pressure**, and **van't Hoff factor** from a 270-minute YouTube video. (source)

Section 1: Solutions Fundamentals and Classification 🧪

This section covers the foundational concepts of solutions, their classification, solubility principles, and Henry's Law from a 223-minute YouTube video (source)

Introduction and Chapter Overview

The Solutions chapter in Physical Chemistry is characterized as a **short, easy chapter** that typically yields **one direct question in JEE Mains**—the type that can be solved quickly upon recognition. The entire chapter focuses exclusively on **binary solutions** (one solute + one solvent), though ternary (three components) and quaternary (four components) solutions exist in advanced contexts.

> **Key Foundational Principle**: "It is the solvent which decides the physical state of the solution." The solvent's physical state determines whether we have gaseous, liquid, or solid solutions.

Classification of Solutions

Solutions are classified into three types based on the **solvent's physical state**:

Type	Solvent State	Solute Can Be	Examples
Gaseous Solutions	Gas	Solid, Liquid, or Gas	Air (mixture of gases)
Liquid Solutions	Liquid	Solid, Liquid, or Gas	Sugar solution, alcohol-water, aerated drinks
Solid Solutions	Solid	Solid, Liquid, or Gas	Brass, bronze (alloys)

Syllabus Focus: The chapter exclusively studies **liquid solutions**, particularly where both solute and solvent are liquids, or where solute is solid/non-volatile.

Solubility Fundamentals

Definition

> **Solubility**: "Maximum amount of solute that can be dissolved in a specified amount of solvent at a particular temperature and pressure."

The key phrase is **"maximum amount"** — this distinguishes saturated solutions from unsaturated ones.

Types of Solutions Based on Saturation

Type	Definition	Characteristics
Unsaturated	More solute can be dissolved at that temperature	Below saturation limit
Saturated	Maximum solute dissolved; no more can dissolve	Dynamic equilibrium exists
Supersaturated	Contains more dissolved solute than normally possible	Unstable; achieved by heating then cooling

Supersaturated Solution Example: Sugar syrup (chashni) used in Indian weddings. When heated, more sugar dissolves; upon cooling, excess sugar crystallizes on the surface.

Equilibrium in Saturated Solutions

In a saturated solution, **dynamic equilibrium** exists:

> **Dissolution rate = Crystallization rate**

Represented as: Solute + Solvent ⇌ Solution

- **Forward arrow**: Dissolution
- **Backward arrow**: Crystallization

Factors Affecting Solubility

1. Nature of Solute and Solvent (Like Dissolves Like)

> **Principle**: "Like dissolves like" — substances with similar intermolecular forces/polarity dissolve in each other.

Solute Type	Solvent Type	Example
Polar	Polar	Sugar in water, HCl in water
Non-polar	Non-polar	Naphthalene in benzene, hydrocarbons in CCl_4

Polar solvents: Water, acetic acid, alcohols **Non-polar solvents**: Benzene, CCl_4, CS_2, hydrocarbons

Gas Solubility Enhancement: Gases that form ions in water are more soluble due to polarity matching:

- $HCl \rightarrow H^+ + Cl^-$
- $NH_3 \rightarrow NH_4^+ + OH^-$
- $CO_2 \rightarrow H_2CO_3 \rightarrow 2H^+ + CO_3^{2-}$

2. Temperature Effect

For Solids in Liquids:

- If dissolution is **endothermic** ($\Delta H > 0$): Solubility **increases** with temperature
- If dissolution is **exothermic** ($\Delta H < 0$): Solubility **decreases** with temperature

> **Application of Le Chatelier's Principle**: Temperature increase shifts equilibrium toward endothermic direction.

For Gases in Liquids:

- Dissolution is always **exothermic** (gas → liquid resembles condensation)
- **Solubility decreases with temperature increase**
- **Environmental consequence**: Rising water temperatures reduce dissolved oxygen, harming aquatic life (anoxia)

3. Pressure Effect

For Solids and Liquids in Liquids: Negligible effect (incompressible)

For Gases in Liquids: **Henry's Law**

Henry's Law

Statement

> "The partial pressure of a gas in vapor phase (p) is directly proportional to the mole fraction of the gas in solution (x)."

Mathematical form: $p = K_H \cdot x$

Where:

- p = partial pressure of gas
- x = mole fraction of gas in solution
- K_H = Henry's constant (depends on nature of gas, temperature)

Key Properties of K_H

Property	Explanation
Inverse solubility relationship	Higher K_H → Lower solubility
Temperature dependence	K_H increases with temperature (solubility decreases)
Gas specificity	Different gases have different K_H values at same temperature

Solvent specificity	Same gas has different K_H in different solvents

Applications of Henry's Law

Application	Explanation
Carbonated drinks (soft drinks)	CO_2 dissolved under high pressure; when bottle opened, pressure drops → CO_2 solubility decreases → bubbles form ("fizz" sound)
Scuba diving ("the bends")	At depth: High pressure increases N_2 and O_2 solubility in blood; rapid ascent: Pressure drops too quickly → gases form bubbles in blood (bends); **Prevention**: Use He-O_2 mixture (helium less soluble, prevents bends)
High altitude sickness (anoxia)	Low atmospheric pressure at altitude; reduced O_2 solubility in blood; symptoms: Nausea, fatigue, confusion, incoherent speech

Limitations of Henry's Law

Henry's Law applies **only when**:

Condition	Explanation
Temperature is not too low	Extreme cold affects gas behavior
Pressure is not too high	Very high pressure causes deviations
Gas does NOT associate or dissociate in solvent	Chemical changes alter particle count
Gas does NOT chemically react with solvent	Reaction changes the dissolved species

Examples where Henry's Law fails:

- HCl in water (dissociates: $HCl \rightarrow H^+ + Cl^-$)
- NH_3 in water (reacts: $NH_3 + H_2O \rightarrow NH_4^+ + OH^-$)

- CO_2 in water (reacts to form H_2CO_3)

Section 2: Vapor Pressure, Raoult's Law, and Solution Thermodynamics 🌡️

This section examines vapor pressure fundamentals, Raoult's Law applications, ideal and non-ideal solution behavior, and azeotropic mixtures from a 393-minute YouTube video (source)

Vapor Pressure of Pure Liquids

Fundamental Definition

> **Vapor Pressure**: The pressure exerted by saturated vapors in equilibrium with the liquid at a particular temperature.

Equilibrium Process Dynamics:

- Liquid molecules at surface possess higher energy
- They escape into vapor phase (evaporation)
- Vapor molecules simultaneously condense back
- Initially: Rate of evaporation = maximum, Rate of condensation = zero
- Eventually: Rate of evaporation = Rate of condensation → **Dynamic equilibrium established**

Critical Insight: Vapor pressure depends **ONLY on temperature** for a particular liquid, **NOT** on:

- Amount of liquid (provided sufficient for equilibrium)
- Shape or size of container
- Volume of container

Mathematical Temperature Relationship

The equilibrium constant Kp for liquid-vapor equilibrium equals vapor pressure:

- **Kp = P_vapor**

Since Kp follows the van't Hoff equation: $\log\left(\frac{K_{p2}}{K_{p1}}\right) = \frac{\Delta H_{vap}}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$

Therefore, vapor pressure has **exponential (not linear) dependence on temperature**.

Key True/False Distinctions:

Statement	Answer	Explanation
"Vapor pressure increases on increasing temperature"	**TRUE**	Higher temperature → more molecules escape
"Vapor pressure increases linearly with temperature"	**FALSE**	Relationship is exponential, not linear

Effect of Intermolecular Forces

Vapor pressure ∝ 1/(Force of attraction)

Order of intermolecular forces:

> **Hydrogen bonding > Dipole-dipole > Van der Waals**

Comparative Example: Water vs Chloroform at same temperature

- Water has **H-bonding** (strongest) → Lower vapor pressure
- Chloroform has **dipole-dipole** → Higher vapor pressure
- **Conclusion**: Chloroform is more volatile than water

For non-polar molecules: Higher molecular mass → Stronger London forces → Lower vapor pressure

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Raoult's Law and Solution Vapor Pressure

Case 1: Non-volatile Solute in Volatile Solvent

> **Raoult's Law**: $P_{solution} = P°_{solvent} \times X_{solvent}$

Where:

- $P°_{solvent}$ = vapor pressure of pure solvent
- $X_{solvent}$ = mole fraction of solvent

Key Consequence: $P_{solution} < P°_{solvent}$ (vapor pressure lowering)

Physical Reasons:

1. Solute-solvent interactions reduce solvent's tendency to evaporate

2. Solute particles occupy surface area, reducing solvent molecules at surface

Case 2: Both Components Volatile (Miscible)

For ideal solution of two volatile liquids A and B:

Total vapor pressure: $P_{total} = P^\circ{}_A \times X_A + P^\circ{}_B \times X_B$

Vapor phase composition (Dalton's Law):

- $Y_A = (P^\circ_A \times X_A) / P_{total}$
- $Y_B = (P^\circ_B \times X_B) / P_{total}$

Alternative Useful Form: $\frac{Y_A}{Y_B} = \frac{P^\circ{}_A}{P^\circ{}_B} \times \frac{X_A}{X_B}$

Case 3: Immiscible Volatile Liquids (Agitated System)

When two immiscible liquids are vigorously agitated:

- Both can evaporate despite not mixing
- Each behaves as if pure ($X = 1$ for both)
- **Total vapor pressure = $P^\circ_A + P^\circ_B$** (simple sum!)

Vapor composition:

- $Y_A = P^\circ_A / (P^\circ_A + P^\circ_B)$
- $Y_B = P^\circ_B / (P^\circ_A + P^\circ_B)$

Mole ratio in vapor: $n_A/n_B = P^\circ_A/P^\circ_B$

Weight ratio: $w_A/w_B = (P^\circ_A \times M_A)/(P^\circ_B \times M_B)$

Ideal and Non-Ideal Solutions

Ideal Solutions

Condition: A-A interactions = B-B interactions = A-B interactions

In other words: Force of attraction between A-A + Force of attraction between B-B = Force of attraction between A-B in solution

Thermodynamic Characteristics:

Property	Value	Interpretation

ΔV_{mixing}	0	No volume change on mixing
ΔH_{mixing}	0	No heat change - neither absorbed nor released
ΔS_{mixing}	> 0	Entropy increases - randomness increases
ΔG_{mixing}	< 0	Process is spontaneous

Examples of Ideal Solutions:

- Benzene + Toluene
- n-Hexane + n-Pentane
- Chlorobenzene + Bromobenzene
- Ethyl chloride + Ethyl bromide
- CCl_4 + $CHCl_3$

> **Key Point**: Perfect ideal solutions don't exist in reality; we approximate with substances having very similar properties.

Non-Ideal Solutions

Definition: Solutions that do **NOT** obey Raoult's Law

- $P_{solution} \neq P°_A \cdot X_A + P°_B \cdot X_B$

Positive Deviation from Raoult's Law

Condition: A-B interactions < A-A or B-B interactions (weaker new interactions)

Vapor Pressure Behavior: $P_{observed} > P_{calculated}$

Thermodynamic Signs:

Property	Sign	Interpretation
ΔH_{mixing}	> 0 (positive)	Endothermic - heat absorbed
ΔV_{mixing}	> 0 (positive)	Volume increases
ΔS_{mixing}	> 0	Randomness increases
ΔG_{mixing}	< 0	Spontaneous

Examples:

- Ethanol + Water
- Acetone + CS_2
- Acetone + CCl_4
- Alcohol + Hydrocarbons
- Cyclohexane + Ethanol

Explanation: Ethanol has H-bonding; when mixed with water, weaker H-bonds form between ethanol-water compared to ethanol-ethanol or water-water, reducing overall interactions.

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Thermodynamic Signs:

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ΔH_mixing	< 0 (negative)	Exothermic - heat released
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Examples:

- HCl + Water
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- Phenol + Aniline
- Chloroform + Acetone
- Acetic acid + Pyridine

Explanation (Chloroform + Acetone): Chloroform's acidic H interacts with acetone's oxygen, forming strong hydrogen bonding, stronger than original dipole-dipole forces.

Explanation (Acetic acid + Pyridine): Pyridine's nitrogen (less electronegative than oxygen) forms stronger H-bond with acetic acid's H than acetic acid's own H-bonding (O-H...O). N-H...N is stronger than O-H...O because nitrogen's lone pair is more available.

Azeotropes (Azeotropic Mixtures)

> **Definition**: Binary liquid mixtures where **liquid and vapor phase compositions are identical** at a particular composition. These cannot be separated by simple or fractional distillation.

Key Characteristic: X_A (liquid) = Y_A (vapor) and X_B (liquid) = Y_B (vapor) at azeotropic composition

Formation: Only **non-ideal solutions** form azeotropes; ideal solutions NEVER form azeotropes

Types of Azeotropes

Type	Deviation	Boiling Point	Vapor Pressure	Example
Minimum Boiling	Positive	**Lower** than both components	Higher than expected	Ethanol-Water (95.6% ethanol, bp 78.2°C)
Maximum Boiling	Negative	**Higher** than both components	Lower than expected	HNO_3-Water (68% HNO_3, bp 108.6°C) or 20.2% HCl at 108.6°C

Key Insight: For azeotropes, Y_A = X_A and Y_B = X_B. The distillation process cannot proceed beyond this point—repeated distillation yields the same composition.

Distillation Concepts

Type of Distillation	Method	Principle
Simple Distillation	Vary temperature at constant pressure	Separate based on boiling point difference
Fractional Distillation	Multiple vaporization-condensation cycles	Enrich more volatile component progressively

Distillation at Reduced Pressure	Vary pressure at constant temperature	Lower boiling points for heat-sensitive substances

For Ideal Solutions: Fractional distillation can completely separate components. The vapor is always richer in more volatile component.

For Non-Ideal Solutions with Azeotropes: Cannot separate beyond azeotropic composition.

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Vapor composition:

- $Y_A = P°_A / (P°_A + P°_B)$
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Mole ratio in vapor: $n_A/n_B = P°_A/P°_B$

Weight ratio: $w_A/w_B = (P°_A \times M_A)/(P°_B \times M_B)$

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Azeotropes (Azeotropic Mixtures)

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Key Characteristic: X_A (liquid) = Y_A (vapor) and X_B (liquid) = Y_B (vapor) at azeotropic composition

Formation:

- **Only non-ideal solutions** form azeotropes
- Ideal solutions NEVER form azeotropes
- Results from **extreme deviations** from Raoult's Law

Types of Azeotropes

Type	Deviation	Boiling Point	Vapor Pressure	Graph Shape
Minimum Boiling Azeotrope	Positive (extreme)	**Lower** than both components	Higher than expected	Leaf-shaped, maximum at top
Maximum Boiling Azeotrope	Negative (extreme)	**Higher** than both components	Lower than expected	Leaf-shaped, minimum at top

Examples from NCERT:

- **Minimum boiling**: Ethanol-Water (95.6% ethanol, boils at 78.2°C vs pure ethanol 78.5°C and water 100°C)
- **Maximum boiling**: HNO_3-Water (68% HNO_3, boils at 121.9°C or 393.5 K)

Graphical Representation

The complete vapor pressure-composition diagram shows:

- **Liquid composition curve**: Straight line for ideal, curved for non-ideal
- **Vapor composition curve**: Always below liquid curve for minimum boiling, above for maximum boiling
- **Azeotropic point**: Where liquid and vapor curves touch (tangent)

At this point, distillation cannot separate components because distillate has same composition as residue.

Distillation Concepts

Types of Distillation

Type	Method	Principle
Simple Distillation	Vary temperature at constant pressure	Separate based on boiling point difference
Fractional Distillation	Multiple vaporization-condensation cycles	Enrich more volatile component progressively
Distillation at Reduced Pressure	Vary pressure at constant temperature	Lower boiling points for heat-sensitive substances

Pressure-Temperature Relationship in Distillation

The instructor emphasizes a **new perspective**: Instead of always heating to boil, we can **reduce pressure** to achieve boiling at constant temperature.

Example from graph analysis:

- At high pressure: Only liquid exists
- At low pressure: Only vapor exists
- Intermediate region: Liquid and vapor in equilibrium

Bubble Point: First bubble of vapor forms when pressure is reduced (point B on diagram)

Dew Point: Last drop of liquid remains (point C on diagram)

Between B and C: **Two-phase region** (liquid + vapor in equilibrium)

Section 3: Colligative Properties, van't Hoff Factor, and JEE Problem-Solving 🎯

This section covers the four colligative properties, van't Hoff factor calculations, Henry's Law applications, and strategic exam preparation from a 270-minute YouTube video (source)

Introduction and Context

Chapter Significance:

- Dilute Solutions is now the **first chapter of Class 12 Physical Chemistry** (previously Solid State held this position)
- Total Physical Chemistry chapters in Class 12 reduced to **three**: (1) Dilute Solutions, (2) Electrochemistry, (3) Chemical Kinetics
- Solid State and Surface Chemistry removed from JEE Main syllabus

Exam Statistics: In the last 6 years, approximately **170 questions** asked from Dilute Solutions in JEE exams. Of these, 50-60 questions overlap with Mole Concept (molality, molarity, mole fraction calculations).

Core Topics Covered:

- Ideal and Non-ideal solutions
- Vapor pressure, Raoult's Law, Dalton's Law of Partial Pressure
- Colligative properties: Relative lowering of vapor pressure, Elevation in boiling point, Depression in freezing point, Osmotic pressure
- van't Hoff factor, Henry's Law
- Azeotropic mixtures

Colligative Properties

Definition and Scope

> **Colligative Properties**: Properties of solutions containing **non-volatile solute in volatile solvent** that depend **only on the number of solute particles**, not on nature of solute.

Key Condition: Solute must be **non-volatile**

Four Colligative Properties:

Property	Symbol	Depends on Solvent Nature?
Relative Lowering of Vapor Pressure	RLVP	NO
Elevation in Boiling Point	ΔT_b	YES
Depression in Freezing Point	ΔT_f	YES
Osmotic Pressure	π	NO

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1. Relative Lowering of Vapor Pressure (RLVP)

Derivation and Formula

For a solution with non-volatile solute B:

- Pure solvent vapor pressure: P°_A
- Solution vapor pressure: P_s = P°_A · X_A (since P_B = 0)

Lowering of vapor pressure: $\Delta p = P^\circ{}_A - P_s = P^\circ{}_A - P^\circ{}_A \cdot X_A = P^\circ{}_A(1 - X_A) = P^\circ{}_A \cdot X_B$

Relative lowering of vapor pressure: $\frac{\Delta p}{P^\circ{}_A} = \frac{P^\circ{}_A - P_s}{P^\circ{}_A} = X_B = \frac{n_B}{n_A + n_B}$

For **dilute solutions**: n_A >> n_B, so: $\frac{P^\circ{}_A - P_s}{P^\circ{}_A} \approx \frac{n_B}{n_A} = \frac{w_B/M_B}{w_A/M_A}$

Alternative Forms

$\frac{P^\circ{}_A - P_s}{P_s} = \frac{n_B}{n_A} = \frac{w_B \cdot M_A}{M_B \cdot w_A}$

Key Distinction:

- If **RLVP is given**: Must use exact formula with n_solute + n_solvent in denominator
- If **P° and P_s are given separately**: Can use (P° - P_s)/P_s = n_solute/n_solvent (easier calculation)

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2. Elevation in Boiling Point (ΔT_b)

Definition

The increase in boiling point when a non-volatile solute is added to a solvent.

$$\Delta T_b = T_b - T_b^0$$

Where:

- T_b^0 = boiling point of pure solvent
- T_b = boiling point of solution

Formula

For dilute solutions: $\Delta T_b = K_b \cdot m$

Where:

- K_b = molal elevation constant (ebullioscopic constant)
- m = molality (mol solute/kg solvent)

Alternative Expression

$$\Delta T_b = \frac{K_b \cdot w_B \cdot 1000}{M_B \cdot w_A}$$

Molal Elevation Constant (K_b)

> **Definition**: Elevation in boiling point when molality of solution is unity (1 mol/kg).

Alternative names:

- Ebullioscopic constant
- Molal boiling point elevation constant

Unit: $K \cdot kg \cdot mol^{-1}$ (or $°C \cdot kg \cdot mol^{-1}$)

Formula relating to thermodynamic quantities: $K_b = \frac{R(T_b^0)^2 \cdot M_A}{1000 \cdot \Delta H_{vap}}$

Or using latent heat per mole: $K_b = \frac{R(T_b^0)^2}{\Delta H_{vap,m}}$

Where:

- R = gas constant (8.314 $J \cdot K^{-1} \cdot mol^{-1}$ or 0.0821 $L \cdot atm \cdot K^{-1} \cdot mol^{-1}$)
- T_b^0 = boiling point of pure solvent (K)
- M_A = molar mass of solvent (g/mol)
- ΔH_vap = latent heat of vaporization per gram
- ΔH_vap, m = molar latent heat of vaporization

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3. Depression in Freezing Point (ΔT_f)

Definition

The decrease in freezing point when a non-volatile solute is added to a solvent.

$$\Delta T_f = T_f^0 - T_f$$

Where:

- T_f^0 = freezing point of pure solvent
- T_f = freezing point of solution

Freezing Point Definition

> "The temperature at which the vapor pressure of a substance in its liquid phase equals the vapor pressure of the solid phase."

At freezing point: Solid ⇌ Liquid (equilibrium)

Formula

$$\Delta T_f = K_f \cdot m = \frac{K_f \cdot w_B \cdot 1000}{M_B \cdot w_A}$$

Molal Depression Constant (K_f)

Alternative names:

- Cryoscopic constant
- Freezing point depression constant
- Molal freezing point depression constant

Unit: K·kg·mol^{-1}

Formula: $K_f = \frac{R(T_f^0)^2 \cdot M_A}{1000 \cdot \Delta H_{fus}}$

Or: $K_f = \frac{R(T_f^0)^2}{\Delta H_{fus,m}}$

Where ΔH_fus = latent heat of fusion.

Practical Application: Ice-Salt Mixture

Adding salt to ice lowers the freezing point, causing ice to melt. Used in:

- Ice cream making

- De-icing roads

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4. Osmotic Pressure (π)

Definitions

> **Osmosis**: The spontaneous flow of solvent molecules from a region of lower concentration (or pure solvent) to a region of higher concentration through a semi-permeable membrane.

> **Osmotic Pressure (π)**: The external pressure that must be applied to the solution side to prevent the net flow of solvent into the solution through a semi-permeable membrane.

Key Characteristics

Aspect	Description
Direction	Pure solvent → Solution (or dilute → concentrated)
Driving force	Concentration gradient
SPM allows	Only solvent molecules
SPM blocks	Solute molecules (larger size)

Types of Solutions Based on Osmotic Pressure

Type	Definition	Osmotic Pressure Comparison
Isotonic	Equal osmotic pressure	$\pi_1 = \pi_2$
Hypertonic	Higher osmotic pressure	$\pi_1 > \pi_2$
Hypotonic	Lower osmotic pressure	$\pi_1 < \pi_2$

Osmosis direction: Hypotonic → Hypertonic (or from lower π to higher π)

Formula

$$\pi = C \cdot R \cdot T = \frac{n}{V} \cdot R \cdot T$$

Where:

- C = concentration (mol/L)
- R = gas constant (0.0821 L·atm·K^{-1}·mol^{-1} for π in atm; 8.314 J·K^{-1}·mol^{-1} for π in Pa)
- T = temperature (K)
- V = volume of solution (L)

For molar mass determination: $\pi = \frac{w_B \cdot R \cdot T}{M_B \cdot V}$

Therefore: $M_B = \frac{w_B \cdot R \cdot T}{\pi \cdot V}$

Reverse Osmosis (RO)

> **Principle**: Applying external pressure **greater than** osmotic pressure to the solution side, forcing solvent to flow from solution to pure solvent side.

Application: Water purification (RO water filters)

- Salt water (impure) → Pressure > π → Pure water separated
- **Limitation**: Removes essential minerals along with impurities

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Van't Hoff Factor (i) and Abnormal Molar Masses

Need for Van't Hoff Factor

When solutes **dissociate** or **associate** in solution, the actual number of particles differs from the calculated value, causing **abnormal colligative properties**.

Phenomenon	Effect on Particle Number	Effect on Colligative Properties
Dissociation	Increases	Observed > Calculated
Association	Decreases	Observed < Calculated

Van't Hoff Factor (i)

> **Definition**: The ratio of observed colligative property to calculated colligative property.

$$i = \frac{\text{Observed colligative property}}{\text{Calculated colligative property}} = \frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$$

For Dissociation

$i = 1 - \alpha + n\alpha$

Where:

- α = degree of dissociation (fraction dissociated)
- n = number of particles formed from one formula unit

Common cases:

Compound	Dissociation	n	i (if α=1)
$NaCl$	$Na^+ + Cl^-$	2	2
K_2SO_4	$2K^+ + SO_4^{2-}$	3	3
$Al_2(SO_4)_3$	$2Al^{3+} + 3SO_4^{2-}$	5	5

Properties:

- i > 1 for dissociation
- If α = 1 (100% dissociation): i = n

For Association (Dimerization)

$i = 1 - \alpha + \frac{\alpha}{n}$

Where:

- α = degree of association
- n = number of molecules associating (typically 2 for dimerization)

Common example: Acetic acid in benzene forms dimers through hydrogen bonding: $2CH_3COOH \rightleftharpoons (CH_3COOH)_2$

Properties:

- i < 1 for association
- If α = 1 (100% dimerization): i = 1/n = 0.5

Modified Colligative Property Formulas

With van't Hoff factor:

Property	Original Formula	Modified Formula

RLVP	$\frac{p_A^0 - p_s}{p_A^0} = x_B$	$\frac{p_A^0 - p_s}{p_A^0} = i \cdot x_B$
Elevation in BP	$\Delta T_b = K_b \cdot m$	$\Delta T_b = i \cdot K_b \cdot m$
Depression in FP	$\Delta T_f = K_f \cdot m$	$\Delta T_f = i \cdot K_f \cdot m$
Osmotic Pressure	$\pi = CRT$	$\pi = iCRT$

Key Problem-Solving Strategies and JEE Questions

Important Relationships to Remember

1. **For water**: Normal boiling point = 100°C = 373 K; Normal freezing point = 0°C = 273 K
2. **At boiling point**: Vapor pressure of liquid = External pressure (1 atm = 760 mmHg = 1.013×10^5 Pa)
3. **Henry's Law constant units**: Same as pressure units (bar, atm, Pa, etc.)
4. **Raoult's Law is a special case of Henry's Law** when K_H = p_A^0

Common JEE Question Types

Question Type	Description
Direct formula application	Given values, calculate molar mass, osmotic pressure, etc.
Graph interpretation	Identifying positive/negative deviation, ideal behavior, azeotropes
van't Hoff factor calculations	Determining degree of dissociation/association
Conceptual questions	Identifying when Henry's Law fails, direction of osmosis, etc.

Sample Problem Structure (Elevation in Boiling Point)

Given: Mass of solute (w_B), mass of solvent (w_A), boiling point of solution (T_b), K_b of solvent, boiling point of pure solvent (T_b^0)

Find: Molar mass of solute (M_B)

Solution steps:

1. Calculate ΔT_b = T_b - T_b^0

2. Use formula: $\Delta T_b = (K_b \cdot w_B \cdot 1000)/(M_B \cdot w_A)$
3. Solve for M_B

Summary of Key Formulas

Topic	Formula	Variables
Henry's Law	$p = K_H \cdot x$	p = partial pressure, K_H = Henry's constant, x = mole fraction in solution
Raoult's Law	$p_A = p_A^0 \cdot x_A$	p_A^0 = pure component vapor pressure, x_A = mole fraction in liquid
Total VP (ideal)	$p_{total} = p_A^0 x_A + p_B^0 x_B$	For binary liquid-liquid solutions
RLVP	$(p_A^0 - p_s)/p_A^0 = x_B \approx n_B/n_A$	For dilute solutions with non-volatile solute
Elevation in BP	$\Delta T_b = K_b \cdot m = (K_b \cdot w_B \cdot 1000)/(M_B \cdot w_A)$	$K_b = R(T_b^0)^2 M_A/(1000\ \Delta H_{vap})$
Depression in FP	$\Delta T_f = K_f \cdot m = (K_f \cdot w_B \cdot 1000)/(M_B \cdot w_A)$	$K_f = R(T_f^0)^2 M_A/(1000\ \Delta H_{fus})$
Osmotic Pressure	$\pi = CRT = nRT/V = w_B RT/(M_B V)$	C = molarity, R = gas constant, T = temperature (K)
van't Hoff Factor	i = observed property/calculated property = $M_{normal}/M_{abnormal}$	For dissociation: $i = 1 + (n-1)\alpha$; For association: $i = 1 - \alpha + \alpha/n$

Final Notes and Study Recommendations

The instructor emphasizes this chapter is **high-scoring for JEE** with **straightforward, formula-based questions**. Key success strategies:

Strategy	Implementation
Master all formulas	Learn variations with van't Hoff factor

Practice unit conversions	Pressure: atm, mmHg, Pa, bar; Temperature: °C to K
Memorize examples	Positive/negative deviation and azeotropes (direct memory required)
Understand graphical representations	Vapor pressure vs. mole fraction for ideal, positive deviation, negative deviation cases
Work through previous JEE questions	1990-2023 to recognize pattern types

The chapter concludes with the van't Hoff factor and abnormal molar masses, which modifies all previous colligative property formulas when solutes dissociate or associate in solution.

Section 3: Colligative Properties, van't Hoff Factor, and Strategic JEE Preparation 📚

This section covers the four colligative properties in depth, van't Hoff factor calculations, Henry's Law applications, and exam preparation strategies from a 270-minute YouTube video (source)

Introduction and Chapter Context

Instructor: Session led for "VCS" batch students for JEE Main/Advanced preparation.

Chapter Significance:

- Dilute Solutions is now the **first chapter of Class 12 Physical Chemistry**
- Total Physical Chemistry chapters reduced to **three**: (1) Dilute Solutions, (2) Electrochemistry, (3) Chemical Kinetics
- Solid State and Surface Chemistry removed from JEE Main syllabus

Exam Statistics: Approximately **170 questions** from Dilute Solutions in last 6 years of JEE exams. Of these, 50-60 overlap with Mole Concept (molality, molarity, mole fraction calculations).

Pedagogical Advice:

- One-shot videos must be followed by **practice of 20-30 questions per topic**
- PYQs (Previous Year Questions) are most important as NTA is predictable

- Focus on understanding which topics have fixed question patterns

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Colligative Properties

Definition

> **Colligative Properties**: Properties of dilute solutions that depend **only on the number of solute particles** per unit volume of solution, not on the nature of solute.

Key Condition: Solute must be **non-volatile**

Four Colligative Properties

Property	Symbol	Definition	Depends on Solvent?
Relative Lowering of Vapor Pressure	RLVP	Ratio of lowering of vapor pressure to pure solvent's vapor pressure	NO
Elevation in Boiling Point	ΔT_b	Increase in boiling point of solution vs. pure solvent	YES
Depression in Freezing Point	ΔT_f	Decrease in freezing point of solution vs. pure solvent	YES
Osmotic Pressure	π	External pressure required to prevent osmosis	NO

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1. Relative Lowering of Vapor Pressure (RLVP)

Derivation and Formula

For a solution with non-volatile solute B:

- Pure solvent vapor pressure: P°_A
- Solution vapor pressure: $P_s = P^\circ_A \cdot X_A$ (since $P_B = 0$)

Lowering of vapor pressure: $\Delta p = P^\circ{}_A - P_s = P^\circ{}_A - P^\circ{}_A \cdot X_A = P^\circ{}_A(1 - X_A) = P^\circ{}_A \cdot X_B$

Relative lowering of vapor pressure: $\frac{\Delta p}{P^\circ{}_A} = \frac{P^\circ{}_A - P_s}{P^\circ{}_A} = X_B = \frac{n_B}{n_A + n_B}$

For **dilute solutions**: n_A >> n_B, so: $\frac{P^\circ{}_A - P_s}{P^\circ{}_A} \approx \frac{n_B}{n_A} = \frac{w_B/M_B}{w_A/M_A}$

Alternative Forms

$\frac{P^\circ{}_A - P_s}{P_s} = \frac{n_B}{n_A} = \frac{w_B \cdot M_A}{M_B \cdot w_A}$

Key Distinction:

- If **RLVP is given**: Must use exact formula with n_solute + n_solvent in denominator
- If **P° and P_s are given separately**: Can use (P° - P_s)/P_s = n_solute/n_solvent (easier calculation)

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2. Elevation in Boiling Point (ΔT_b)

Definition

The increase in boiling point when a non-volatile solute is added to a solvent.

$$\Delta T_b = T_b - T_b^0$$

Where:

- T_b^0 = boiling point of pure solvent
- T_b = boiling point of solution

Formula

For dilute solutions: $\Delta T_b = K_b \cdot m$

Where:

- K_b = molal elevation constant (ebullioscopic constant)
- m = molality (mol solute/kg solvent)

Alternative Expression

$$\Delta T_b = \frac{K_b \cdot w_B \cdot 1000}{M_B \cdot w_A}$$

Molal Elevation Constant (K_b)

> **Definition**: Elevation in boiling point when molality of solution is unity (1 mol/kg).

Alternative names:

- Ebullioscopic constant
- Molal boiling point elevation constant

Unit: $K \cdot kg \cdot mol^{-1}$ (or $°C \cdot kg \cdot mol^{-1}$)

Formula relating to thermodynamic quantities: $K_b = \frac{R(T_b^0)^2 \cdot M_A}{1000 \cdot \Delta H_{vap}}$

Or using latent heat per mole: $K_b = \frac{R(T_b^0)^2}{\Delta H_{vap,m}}$

Where:

- R = gas constant (8.314 $J \cdot K^{-1} \cdot mol^{-1}$ or 0.0821 $L \cdot atm \cdot K^{-1} \cdot mol^{-1}$)
- T_b^0 = boiling point of pure solvent (K)
- M_A = molar mass of solvent (g/mol)
- ΔH_vap = latent heat of vaporization per gram
- ΔH_vap, m = molar latent heat of vaporization

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3. Depression in Freezing Point (ΔT_f)

Definition

The decrease in freezing point when a non-volatile solute is added to a solvent.

$$\Delta T_f = T_f^0 - T_f$$

Where:

- T_f^0 = freezing point of pure solvent
- T_f = freezing point of solution

Freezing Point Definition

> "The temperature at which the vapor pressure of a substance in its liquid phase equals the vapor pressure of the solid phase."

At freezing point: Solid ⇌ Liquid (equilibrium)

Formula

$\Delta T_f = K_f \cdot m = \frac{K_f \cdot w_B \cdot 1000}{M_B \cdot w_A}$

Molal Depression Constant (K_f)

Alternative names:

- Cryoscopic constant
- Freezing point depression constant
- Molal freezing point depression constant

Unit: $K \cdot kg \cdot mol^{-1}$

Formula: $K_f = \frac{R(T_f^0)^2 \cdot M_A}{1000 \cdot \Delta H_{fus}}$

Or: $K_f = \frac{R(T_f^0)^2}{\Delta H_{fus,m}}$

Where ΔH_fus = latent heat of fusion.

Practical Application: Ice-Salt Mixture

Adding salt to ice lowers the freezing point, causing ice to melt. Used in:

- Ice cream making
- De-icing roads

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Key Characteristics

Aspect	Description

Direction	Pure solvent → Solution (or dilute → concentrated)
Driving force	Concentration gradient
SPM allows	Only solvent molecules
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Types of Solutions Based on Osmotic Pressure

Type	Definition	Osmotic Pressure Comparison
Isotonic	Equal osmotic pressure	$\pi_1 = \pi_2$
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Osmosis direction: Hypotonic → Hypertonic (or from lower π to higher π)

Formula

$\pi = C \cdot R \cdot T = \frac{n}{V} \cdot R \cdot T$

Where:

- C = concentration (mol/L)
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- V = volume of solution (L)

For molar mass determination: $\pi = \frac{w_B \cdot R \cdot T}{M_B \cdot V}$

Therefore: $M_B = \frac{w_B \cdot R \cdot T}{\pi \cdot V}$

Reverse Osmosis (RO)

> **Principle**: Applying external pressure **greater than** osmotic pressure to the solution side, forcing solvent to flow from solution to pure solvent side.

Application: Water purification (RO water filters)

- Salt water (impure) → Pressure > π → Pure water separated
- **Limitation**: Removes essential minerals along with impurities

---

Van't Hoff Factor (i) and Abnormal Molar Masses

Need for Van't Hoff Factor

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$$i = \frac{\text{Observed colligative property}}{\text{Calculated colligative property}} = \frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$$

For Dissociation

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Where:

- α = degree of dissociation (fraction dissociated)
- n = number of particles formed from one formula unit

Common cases:

Compound	Dissociation	n	i (if α=1)
$NaCl$	$Na^+ + Cl^-$	2	2
K_2SO_4	$2K^+ + SO_4^{2-}$	3	3
$Al_2(SO_4)_3$	$2Al^{3+} + 3SO_4^{2-}$	5	5

Properties:

- i > 1 for dissociation
- If α = 1 (100% dissociation): i = n

For Association (Dimerization)

$i = 1 - \alpha + \frac{\alpha}{n}$

Where:

- α = degree of association
- n = number of molecules associating (typically 2 for dimerization)

Common example: Acetic acid in benzene forms dimers through hydrogen bonding: $2CH_3COOH \rightleftharpoons (CH_3COOH)_2$

Properties:

- i < 1 for association
- If α = 1 (100% dimerization): i = 1/n = 0.5

Modified Colligative Property Formulas with 'i'

Property	Original Formula	Modified Formula with i
Relative lowering of VP	(P°-P)/P° = X_solute	(P°-P)/P° = i·n_solute/(n_solute + n_solvent) ≈ i·n_solute/n_solvent
Elevation in boiling point	ΔT_b = K_b·m	ΔT_b = i·K_b·m
Depression in freezing point	ΔT_f = K_f·m	ΔT_f = i·K_f·m
Osmotic pressure	π = CRT	π = i·CRT

Henry's Law

Statement

> **Henry's Law**: At constant temperature, the mole fraction of a gas dissolved in a liquid is directly proportional to the partial pressure of that gas above the liquid.

$X_{gas} \propto P_{gas}$

Or: $P_{gas} = K_H \cdot X_{gas}$

Where:

- X_gas = mole fraction of gas in liquid
- P_gas = partial pressure of gas above liquid
- K_H = Henry's Law constant

Important: The constant in Henry's Law is **K_H** (Henry's constant), NOT the proportionality constant from the first form. When written as P = K_H · X, K_H has units of pressure.

Alternative Form: Sometimes written as: $X_{gas} = \frac{P_{gas}}{K_H}$

Here, larger K_H means lower solubility (inverse relationship).

Factors Affecting Gas Solubility

From Henry's Law and thermodynamics:

Factor	Effect on Solubility	Explanation
High pressure	Increases	Forces gas into liquid
Low temperature	Increases	Gas molecules have less kinetic energy to escape
Nature of gas	Varies	Gases with higher K_H are less soluble
Nature of solvent	Varies	Chemical affinity (e.g., NH_3 in H_2O due to H-bonding)

Practical Example: Carbonated drinks (soda) are bottled under high CO_2 pressure. When opened, pressure drops, CO_2 solubility decreases, and gas bubbles out.

Temperature Effect on Gas Solubility

- **Gas dissolution is exothermic** (similar to condensation)
- By Le Chatelier's principle: **Increasing T decreases gas solubility**
- Therefore: **K_H increases with temperature**

Practical example: Cold water can hold more dissolved O_2 than warm water → aquatic life prefers cold water; fish die in warm water due to oxygen depletion.

Comparison of Gases

Polar gases (e.g., SO_2, NH_3, HCl):

- More soluble in polar solvents (water)
- Lower K_H values

Non-polar gases (e.g., H_2, N_2, O_2):

- Less soluble in polar solvents
- Higher K_H values

Among non-polar gases: Higher molecular mass → stronger London forces → more soluble → lower K_H

Example comparison:

Gas	Nature	Relative K_H	Relative Solubility
SO_2	Polar	Low	High
H_2	Non-polar, low M	High	Low
CO_2	Polar	Medium	Medium

Azeotropic Mixtures

Definition

> **Azeotropes**: Binary liquid mixtures of two volatile components where **liquid and vapor phase compositions are identical** at a particular composition. These cannot be separated by simple distillation or fractional distillation.

Key Characteristic: X_A (liquid) = Y_A (vapor) and X_B (liquid) = Y_B (vapor) at azeotropic composition

Formation:

- **Only non-ideal solutions** form azeotropes
- Ideal solutions NEVER form azeotropes
- Results from **extreme deviations** from Raoult's Law

Types of Azeotropes

Type	Deviation	Boiling Point	Vapor Pressure	Example

Minimum Boiling Azeotrope	Positive (extreme)	**Lower** than both components	Higher than expected	Ethanol-Water (95.6% ethanol, bp 78.2°C)
Maximum Boiling Azeotrope	Negative (extreme)	**Higher** than both components	Lower than expected	HNO_3-Water (68% HNO_3, bp 121.9°C or 393.5 K) or 20.2% HCl at 108.6°C

Key Insight: For azeotropes, Y_A = X_A and Y_B = X_B. The distillation process cannot proceed beyond this point—repeated distillation yields the same composition.

Graphical Representation

The complete vapor pressure-composition diagram shows:

- **Liquid composition curve**: Straight line for ideal, curved for non-ideal
- **Vapor composition curve**: Always below liquid curve for minimum boiling, above for maximum boiling
- **Azeotropic point**: Where liquid and vapor curves touch (tangent)

At this point, distillation cannot separate components because distillate has same composition as residue.

Distillation Concepts

Types of Distillation

Type	Method	Principle
Simple Distillation	Vary temperature at constant pressure	Separate based on boiling point difference
Fractional Distillation	Multiple vaporization-condensation cycles	Enrich more volatile component progressively
Distillation at Reduced Pressure	Vary pressure at constant temperature	Lower boiling points for heat-sensitive substances

Pressure-Temperature Relationship in Distillation

The instructor emphasizes a **new perspective**: Instead of always heating to boil, we can **reduce pressure** to achieve boiling at constant temperature.

Example from graph analysis:

- At high pressure: Only liquid exists
- At low pressure: Only vapor exists
- Intermediate region: Liquid and vapor in equilibrium

Bubble Point: First bubble of vapor forms when pressure is reduced (point B on diagram)

Dew Point: Last drop of liquid remains (point C on diagram)

Between B and C: **Two-phase region** (liquid + vapor in equilibrium)

Boiling Point and Pressure Cooker Logic

Boiling Point Definition

> **Boiling Point**: The temperature at which vapor pressure of liquid equals external (atmospheric) pressure.

Mathematical: At $T = T_b$, $P_{vapor} = P_{external} = 1$ atm (typically)

Notation: T_b° = boiling point of pure solvent

Pressure Cooker Logic (Mountain/High Altitude Cooking)

Situation	Problem	Solution
High altitude	External pressure < 1 atm (e.g., 0.78 atm)	Water boils at lower temperature (e.g., ~90°C instead of 100°C)
Consequence	Food remains undercooked	Insufficient heat/energy for proper cooking
Pressure cooker solution	Increases internal pressure	When internal pressure reaches ~1 atm, water boils at normal temperature
Mechanism	Steam releases through whistle	Maintains pressure, food gets sufficient time and temperature

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Detailed Problem-Solving Examples

Raoult's Law Problem (JEE Main 2024)

Question: Vapor pressures of pure benzene ($P°_A$) = 80 torr, pure methylbenzene/toluene ($P°_B$) = 24 torr at 300K. Find mole fraction of methylbenzene in vapor phase if equimolar mixture in liquid phase forms ideal solution.

Solution:

- Given: $X_A = X_B = 0.5$ (equimolar)
- $P_{total} = P°_A \cdot X_A + P°_B \cdot X_B = 80 \times 0.5 + 24 \times 0.5 = 40 + 12 = 52$ torr

Method 1 - Using Dalton's Law:

- $Y_B = P_B/P_{total} = P°_B \cdot X_B/P_{total} = 24 \times 0.5/52 = 12/52 = 0.23$

Method 2 - Shortcut Equation:

- $P_{total} = Y_A \cdot P°_A + Y_B \cdot P°_B$
- With $Y_A + Y_B = 1$, solve simultaneously

Answer: $Y_B = 0.23 = 23 \times 10^{-2}$

Colligative Property with van't Hoff Factor (JEE Main 2024)

Question: Mass of urea (NH_2CONH_2, M = 60 g/mol) required to be dissolved in 1000 g water so that water's vapor pressure is reduced by 25%. (Given: $P°_{water}$ = 24 mm Hg, find x where final vapor pressure = $x \times 10^{-2}$ mm Hg)

Solution:

- Relative lowering = 25% = 0.25
- For dilute solution: $(P°-P)/P° \approx n_{solute}/n_{solvent}$
- $0.25 = n_{urea} / (1000/18) = n_{urea} / 55.56$
- $n_{urea} = 0.25 \times 55.56 = 13.89$ mol
- Mass = $13.89 \times 60 = 833.4$ g

Note: The answer mentioned in lecture was 111 g for a variant of this problem.

Complex van't Hoff Problem (JEE Main 2021 - Homework)

Question: In a solvent, 50% of an acid A dimerizes ($2A \rightarrow A_2$) and rest dissociates ($A \rightarrow B^+ + C^-$). Find van't Hoff factor.

Solution:

- Let initial A = 1 mol
- 50% dimerizes: 0.5 mol A forms 0.5/2 = 0.25 mol A_2
- Remaining 0.5 mol dissociates: forms 0.5 mol B^+ + 0.5 mol C^-
- Total particles = 0.25 + 0.5 + 0.5 = 1.25
- i = 1.25/1 = 1.25 = **5/4**

Henry's Law Problem (JEE Main 2022)

Question: If O_2 gas bubbled in water at 303 K, find mmol of O_2 dissolved. Given: K_H = 46.82 kbar, P_{O_2} = 0.92 bar, assume very small solubility.

Solution:

- Henry's Law: $P_{gas} = K_H \cdot X_{gas}$
- 0.92 bar = 46.82×10^3 bar × X_{O_2}
- $X_{O_2} = 0.92/(46.82 \times 10^3) = 1.964 \times 10^{-5}$

Since solubility is very small:

- $X_{O_2} \approx n_{O_2} / n_{water}$
- n_{water} = 1000 g / 18 g/mol = 55.56 mol (assuming 1 L water)
- $n_{O_2} = X_{O_2} \times n_{water} = 1.964 \times 10^{-5} \times 55.56 = 1.09 \times 10^{-3}$ mol = **1.09 mmol ≈ 1 mmol**

Osmotic Pressure of Multiple Solutes

For solution with multiple solutes: $\pi_{total} = \pi_1 + \pi_2 + \pi_3 + ... = i_1C_1RT + i_2C_2RT + ...$

Or: $\pi_{total} = RT \times \Sigma(iC)$

Hypertonic, Hypotonic, Isotonic Solutions

Term	Condition	Result
Hypertonic	$\pi_1 > \pi_2$ (or $C_1T_1 > C_2T_2$)	Solvent flows from 2 to 1
Hypotonic	$\pi_1 < \pi_2$ (or $C_1T_1 < C_2T_2$)	Solvent flows from 1 to 2
Isotonic	$\pi_1 = \pi_2$ (or $C_1T_1 = C_2T_2$)	No net flow

At same temperature: Compare concentrations directly

Reverse Osmosis

When external pressure **exceeds** osmotic pressure, solvent flows from concentrated to dilute solution (opposite of osmosis).

Application: Water purification (RO systems), desalination of seawater.

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van't Hoff Factor (i)

Definition

$i = \frac{\text{Actual number of solute particles in solution}}{\text{Original number of solute particles dissolved}}$

Cases

Case 1: Dissociation (Electrolytes)

When solute breaks into multiple particles:

For: A → n particles

$$i = 1 + (n - 1)\alpha$$

Where:

- n = number of particles formed from one formula unit
- α = degree of dissociation (fraction dissociated)

Derivation:

- Initial: 1 mol A
- At equilibrium: (1-α) mol A + nα mol particles
- Total particles = 1 - α + nα = 1 + (n-1)α
- i = [1 + (n-1)α]/1 = 1 + (n-1)α

Case 2: Association (Non-electrolytes, special cases)

When solute particles combine:

For: nA → A_n (dimer, trimer, etc.)

$$i = 1 + \left(\frac{1}{n} - 1\right)\alpha = 1 - \alpha\left(1 - \frac{1}{n}\right)$$

Or: $i = 1 - \alpha + \frac{\alpha}{n}$

Derivation:

- Initial: 1 mol A
- At equilibrium: $(1-\alpha)$ mol A + (α/n) mol A_n
- Total particles = $1 - \alpha + \alpha/n = 1 - \alpha(1 - 1/n)$
- $i = 1 - \alpha(1 - 1/n)$

Modified Colligative Property Formulas with 'i'

Property	Original Formula	Modified Formula with i
Relative lowering of VP	$(P^\circ-P)/P^\circ = X_{solute}$	$(P^\circ-P)/P^\circ = i \cdot n_{solute}/(n_{solute} + n_{solvent}) \approx i \cdot n_{solute}/n_{solvent}$
Elevation in boiling point	$\Delta T_b = K_b \cdot m$	$\Delta T_b = i \cdot K_b \cdot m$
Depression in freezing point	$\Delta T_f = K_f \cdot m$	$\Delta T_f = i \cdot K_f \cdot m$
Osmotic pressure	$\pi = CRT$	$\pi = i \cdot CRT$

Henry's Law

Statement

> **Henry's Law**: At constant temperature, the solubility of a gas in a liquid is directly proportional to the partial pressure of the gas above the liquid.

Mathematical form: $P_{gas} = K_H \times X_{gas}$

Or equivalently: **$X_{gas} = P_{gas}/K_H$**

Where:

- X_{gas} = mole fraction of dissolved gas
- P_{gas} = partial pressure of gas above liquid
- K_H = Henry's law constant

Critical Relationship: K_H is **inversely proportional** to solubility

- High K_H → Low solubility
- Low K_H → High solubility

Temperature Effect on Gas Solubility

- **Gas dissolution is exothermic** (similar to condensation)
- By Le Chatelier's principle: **Increasing T decreases gas solubility**
- Therefore: **K_H increases with temperature**

Practical example: Cold water can hold more dissolved O_2 than warm water → aquatic life prefers cold water; fish die in warm water due to oxygen depletion.

Comparison of Gases

Polar gases (e.g., SO_2, NH_3, HCl):

- More soluble in polar solvents (water)
- Lower K_H values

Non-polar gases (e.g., H_2, N_2, O_2):

- Less soluble in polar solvents
- Higher K_H values

Among non-polar gases: Higher molecular mass → stronger London forces → more soluble → lower K_H

Osmotic Pressure of Multiple Solutes

For solution with multiple solutes: $\pi_{total} = \pi_1 + \pi_2 + \pi_3 + ... = i_1C_1RT + i_2C_2RT + ...$

Or: $\pi_{total} = RT \times \Sigma(iC)$

Hypertonic, Hypotonic, Isotonic Solutions

Term	Condition	Result
Hypertonic	$\pi_1 > \pi_2$ (or $C_1T_1 > C_2T_2$)	Solvent flows from 2 to 1
Hypotonic	$\pi_1 < \pi_2$ (or $C_1T_1 < C_2T_2$)	Solvent flows from 1 to 2
Isotonic	$\pi_1 = \pi_2$ (or $C_1T_1 = C_2T_2$)	No net flow

At same temperature: Compare concentrations directly

Reverse Osmosis

When external pressure **exceeds** osmotic pressure, solvent flows from concentrated to dilute solution (opposite of osmosis).

Application: Water purification (RO systems), desalination of seawater.

Strategic JEE Preparation Advice

Mock Test Strategy (Critical)

Current Status: Two tests completed in VOTS Online Test Series (All India Test Series)

- 15 full syllabus mock tests available
- First mock was free for all
- Second mock on December 12
- **Next mock: December 15**

Success Story: Student improved from 105 to 152 marks between Mock 1 and Mock 2 by:

1. Taking 3-hour mock seriously
2. Spending 1 hour on analysis (identifying weak chapters/topics)
3. Watching video solutions to learn solving techniques
4. Re-attempting questions after understanding the method
5. Reading theory of weak chapters
6. Taking chapter tests (70+ available with solutions)

> **Mandatory Rule**: "Mock test dena zaroori hai" - Taking mock tests is essential, regardless of preparation level. Use any free online test series if VOTS unavailable.

Time Management Strategy (December-January Critical Period)

Current Challenge: ~40 days remaining, cannot spend 20-25 hours per chapter

Smart Approach:

Strategy	Implementation
Complete syllabus first	Focus on connected topics that enable understanding of subsequent topics
Start mocks immediately	Don't wait for complete syllabus
Use approximation in calculations	Allowed in colligative properties (except relative lowering of VP where precision matters)

Follow the cycle	Mock → Video Solution → Theory (if weak) → Chapter Test (10-12 questions) → Next Mock

Approximation Examples:

- 0.792 → 0.8 acceptable
- Range-based answers accepted in JEE

Content-Specific Advice

Topic	Advice
Dilute Solutions	First 2-2.5 hours are difficult (combines JEE Main + Advanced); once foundation set, remaining topics flow easily; last 1 hour (van't Hoff factor + combined problems) most important
P-Block	Complete Chemical Bonding first, then P-block - topics are interrelated
Organic Chemistry	Starting after completing three Physical Chemistry chapters

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Key Homework Assignments

Assignment	Details
NCERT Reading	Read the azeotrope paragraph in NCERT Dilute Solutions chapter; find examples of maximum boiling and minimum boiling azeotropes; write HNO_3 + H_2O example in notes
Problem	2021 JEE Main question on mixed dissociation/association (50% dimerizes, rest dissociates) - solve completely
Previous Year Questions	Complete all unsolved PYQs shown in lecture

Important Values to Remember

Constant	Value	Units
K_b (water)	0.51	$K \cdot kg \cdot mol^{-1}$
K_f (water)	1.86	$K \cdot kg \cdot mol^{-1}$
R	0.0821	$L \cdot atm \cdot K^{-1} \cdot mol^{-1}$
R	8.314	$J \cdot K^{-1} \cdot mol^{-1}$

Final Concept: Solubility of Gases in Liquids

Henry's Law Applications

Application	Explanation
Carbonated beverages	CO_2 dissolved under pressure; when opened, pressure drops → bubbles form
Deep sea diving ("the bends")	N_2 dissolves in blood at high pressure; rapid ascent → bubbles form in blood
Aquatic life	O_2 solubility in water determines fish survival

Temperature Effect

> Gas solubility **decreases** with increasing temperature (explains why warm soda goes flat faster)

Pressure Effect

> Gas solubility **increases** with increasing pressure (Henry's Law)

Section 3: Comprehensive Colligative Properties and JEE Strategy 🎯

This section provides exhaustive coverage of colligative properties, van't Hoff factor applications, Henry's Law, and strategic exam preparation from a 270-minute YouTube video (source)

Introduction and Chapter Context

Instructor: Session led for "VCS" batch students for JEE Main/Advanced preparation.

Chapter Significance:

- Dilute Solutions is now the **first chapter of Class 12 Physical Chemistry**
- Total Physical Chemistry chapters reduced to **three**: (1) Dilute Solutions, (2) Electrochemistry, (3) Chemical Kinetics
- Solid State and Surface Chemistry removed from JEE Main syllabus

Exam Statistics: Approximately **170 questions** from Dilute Solutions in last 6 years of JEE exams. Of these, 50-60 overlap with Mole Concept (molality, molarity, mole fraction calculations).

Pedagogical Advice:

- One-shot videos must be followed by **practice of 20-30 questions per topic**
- PYQs (Previous Year Questions) are most important as NTA is predictable
- Focus on understanding which topics have fixed question patterns

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Colligative Properties

Definition and Scope

> **Colligative Properties**: Properties of dilute solutions that depend **only on the number of solute particles** per unit volume of solution, not on the nature of solute.

Key Condition: Solute must be **non-volatile**

Four Colligative Properties

Property	Symbol	Definition	Depends on Solvent?
Relative Lowering of Vapor Pressure	RLVP	Ratio of lowering of vapor pressure to pure solvent's vapor pressure	NO
Elevation in Boiling Point	ΔT_b	Increase in boiling point of solution vs. pure solvent	YES

Depression in Freezing Point	ΔT_f	Decrease in freezing point of solution vs. pure solvent	YES
Osmotic Pressure	π	External pressure required to prevent osmosis	NO

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1. Relative Lowering of Vapor Pressure (RLVP)

Derivation and Formula

For a solution with non-volatile solute B:

- Pure solvent vapor pressure: P°_A
- Solution vapor pressure: P_s = P°_A · X_A (since P_B = 0)

Lowering of vapor pressure: $\Delta p = P^\circ{}_A - P_s = P^\circ{}_A - P^\circ{}_A \cdot X_A = P^\circ{}_A(1 - X_A) = P^\circ{}_A \cdot X_B$

Relative lowering of vapor pressure: $\frac{\Delta p}{P^\circ{}_A} = \frac{P^\circ{}_A - P_s}{P^\circ{}_A} = X_B = \frac{n_B}{n_A + n_B}$

For **dilute solutions**: n_A >> n_B, so: $\frac{P^\circ{}_A - P_s}{P^\circ{}_A} \approx \frac{n_B}{n_A} = \frac{w_B/M_B}{w_A/M_A}$

Alternative Forms

$\frac{P^\circ{}_A - P_s}{P_s} = \frac{n_B}{n_A} = \frac{w_B \cdot M_A}{M_B \cdot w_A}$

Key Distinction:

- If **RLVP is given**: Must use exact formula with n_solute + n_solvent in denominator
- If **P° and P_s are given separately**: Can use (P° - P_s)/P_s = n_solute/n_solvent (easier calculation)

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2. Elevation in Boiling Point (ΔT_b)

Definition

The increase in boiling point when a non-volatile solute is added to a solvent.

$\Delta T_b = T_b - T_b^0$

Where:

- T_b^0 = boiling point of pure solvent
- T_b = boiling point of solution

Formula

For dilute solutions: $\Delta T_b = K_b \cdot m$

Where:

- K_b = molal elevation constant (ebullioscopic constant)
- m = molality (mol solute/kg solvent)

Alternative Expression

$\Delta T_b = \frac{K_b \cdot w_B \cdot 1000}{M_B \cdot w_A}$

Molal Elevation Constant (K_b)

> **Definition**: Elevation in boiling point when molality of solution is unity (1 mol/kg).

Alternative names:

- Ebullioscopic constant
- Molal boiling point elevation constant

Unit: $K \cdot kg \cdot mol^{-1}$ (or $°C \cdot kg \cdot mol^{-1}$)

Formula relating to thermodynamic quantities: $K_b = \frac{R(T_b^0)^2 \cdot M_A}{1000 \cdot \Delta H_{vap}}$

Or using latent heat per mole: $K_b = \frac{R(T_b^0)^2}{\Delta H_{vap,m}}$

Where:

- R = gas constant ($8.314\ J \cdot K^{-1} \cdot mol^{-1}$ or $0.0821\ L \cdot atm \cdot K^{-1} \cdot mol^{-1}$)
- T_b^0 = boiling point of pure solvent (K)
- M_A = molar mass of solvent (g/mol)
- ΔH_vap = latent heat of vaporization per gram
- ΔH_vap, m = molar latent heat of vaporization

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3. Depression in Freezing Point (ΔT_f)

Definition

The decrease in freezing point when a non-volatile solute is added to a solvent.

$$\Delta T_f = T_f^0 - T_f$$

Where:

- T_f^0 = freezing point of pure solvent
- T_f = freezing point of solution

Freezing Point Definition

> "The temperature at which the vapor pressure of a substance in its liquid phase equals the vapor pressure of the solid phase."

At freezing point: Solid ⇌ Liquid (equilibrium)

Formula

$$\Delta T_f = K_f \cdot m = \frac{K_f \cdot w_B \cdot 1000}{M_B \cdot w_A}$$

Molal Depression Constant (K_f)

Alternative names:

- Cryoscopic constant
- Freezing point depression constant
- Molal freezing point depression constant

Unit: K·kg·mol^{-1}

Formula: $K_f = \frac{R(T_f^0)^2 \cdot M_A}{1000 \cdot \Delta H_{fus}}$

Or: $K_f = \frac{R(T_f^0)^2}{\Delta H_{fus,m}}$

Where ΔH_fus = latent heat of fusion.

Practical Application: Ice-Salt Mixture

Adding salt to ice lowers the freezing point, causing ice to melt. Used in:

- Ice cream making
- De-icing roads

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4. Osmotic Pressure (π)

Definitions

> **Osmosis**: The spontaneous flow of solvent molecules from a region of lower concentration (or pure solvent) to a region of higher concentration through a semi-permeable membrane.

> **Osmotic Pressure (π)**: The external pressure that must be applied to the solution side to prevent the net flow of solvent into the solution through a semi-permeable membrane.

Key Characteristics

Aspect	Description
Direction	Pure solvent → Solution (or dilute → concentrated)
Driving force	Concentration gradient
SPM allows	Only solvent molecules
SPM blocks	Solute molecules (larger size)

Types of Solutions Based on Osmotic Pressure

Type	Definition	Osmotic Pressure Comparison
Isotonic	Equal osmotic pressure	$\pi_1 = \pi_2$
Hypertonic	Higher osmotic pressure	$\pi_1 > \pi_2$
Hypotonic	Lower osmotic pressure	$\pi_1 < \pi_2$

Osmosis direction: Hypotonic → Hypertonic (or from lower π to higher π)

Formula

$$\pi = C \cdot R \cdot T = \frac{n}{V} \cdot R \cdot T$$

Where:

- C = concentration (mol/L)
- R = gas constant (0.0821 $L \cdot atm \cdot K^{-1} \cdot mol^{-1}$ for π in atm; 8.314 $J \cdot K^{-1} \cdot mol^{-1}$ for π in Pa)

- T = temperature (K)
- V = volume of solution (L)

For molar mass determination: $\pi = \frac{w_B \cdot R \cdot T}{M_B \cdot V}$

Therefore: $M_B = \frac{w_B \cdot R \cdot T}{\pi \cdot V}$

Reverse Osmosis (RO)

> **Principle**: Applying external pressure **greater than** osmotic pressure to the solution side, forcing solvent to flow from solution to pure solvent side.

Application: Water purification (RO water filters)

- Salt water (impure) → Pressure > π → Pure water separated
- **Limitation**: Removes essential minerals along with impurities

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van't Hoff Factor (i) and Abnormal Molar Masses

Need for Van't Hoff Factor

When solutes **dissociate** or **associate** in solution, the actual number of particles differs from the calculated value, causing **abnormal colligative properties**.

Phenomenon	Effect on Particle Number	Effect on Colligative Properties
Dissociation	Increases	Observed > Calculated
Association	Decreases	Observed < Calculated

Van't Hoff Factor (i)

> **Definition**: The ratio of observed colligative property to calculated colligative property.

$i = \frac{\text{Observed colligative property}}{\text{Calculated colligative property}} = \frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$

For Dissociation

$i = 1 - \alpha + n\alpha$

Where:

- α = degree of dissociation (fraction dissociated)
- n = number of particles formed from one formula unit

Common cases:

Compound	Dissociation	n	i (if α=1)
NaCl	$Na^+ + Cl^-$	2	2
K_2SO_4	$2K^+ + SO_4^{2-}$	3	3
$Al_2(SO_4)_3$	$2Al^{3+} + 3SO_4^{2-}$	5	5

Properties:

- i > 1 for dissociation
- If α = 1 (100% dissociation): i = n

For Association (Dimerization)

$i = 1 - \alpha + \frac{\alpha}{n}$

Where:

- α = degree of association
- n = number of molecules associating (typically 2 for dimerization)

Common example: Acetic acid in benzene forms dimers through hydrogen bonding: $2CH_3COOH \rightleftharpoons (CH_3COOH)_2$

Properties:

- i < 1 for association
- If α = 1 (100% dimerization): i = 1/n = 0.5

Modified Colligative Property Formulas with 'i'

Property	Original Formula	Modified Formula with i
Relative lowering of VP	$(P^\circ-P)/P^\circ = X_{solute}$	$(P^\circ-P)/P^\circ = i \cdot n_{solute}/(n_{solute} + n_{solvent}) \approx i \cdot n_{solute}/n_{solvent}$
Elevation in boiling point	$\Delta T_b = K_b \cdot m$	$\Delta T_b = i \cdot K_b \cdot m$

Depression in freezing point	$\Delta T_f = K_f \cdot m$	$\Delta T_f = i \cdot K_f \cdot m$
Osmotic pressure	$\pi = CRT$	$\pi = i \cdot CRT$

Strategic JEE Preparation

Mock Test Strategy

Element	Details
Current status	Two tests completed in VOTS Online Test Series
Available	15 full syllabus mock tests
Next mock	December 15
Success pattern	Mock → Video Solution → Theory (if weak) → Chapter Test → Next Mock

Success Story: Student improved from 105 to 152 marks by:

1. Taking 3-hour mock seriously
2. Spending 1 hour on analysis
3. Watching video solutions
4. Re-attempting questions
5. Reading theory of weak chapters
6. Taking chapter tests

> **Mandatory Rule**: "Mock test dena zaroori hai" - Taking mock tests is essential, regardless of preparation level.

Time Management (December-January)

Challenge	Solution
~40 days remaining	Cannot spend 20-25 hours per chapter
Complete syllabus	Focus on connected topics
Start mocks	Don't wait for complete syllabus

Calculations	Use approximation (0.792 → 0.8 acceptable)

Smart Cycle: Mock → Video Solution → Theory (if weak) → Chapter Test (10-12 questions) → Next Mock

Content-Specific Priorities

Topic	Priority	Notes
Dilute Solutions - first 2.5 hours	High	Difficult, combines JEE Main + Advanced
Dilute Solutions - last 1 hour	Critical	van't Hoff factor + combined problems most important
P-Block	Medium	Complete Chemical Bonding first
Organic Chemistry	Starting next week	After three Physical Chemistry chapters

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Key Homework and Values

Homework Assignments

Assignment	Details
NCERT Reading	Azeotrope paragraph; find maximum and minimum boiling azeotrope examples; write $HNO_3 + H_2O$ example
Problem	2021 JEE Main: 50% dimerizes, rest dissociates - find i
PYQs	Complete all unsolved questions from lecture

Important Constants

Constant	Value	Units
K_b (water)	0.51	$K \cdot kg \cdot mol^{-1}$

K_f (water)	1.86	$K \cdot kg \cdot mol^{-1}$
R	0.0821	$L \cdot atm \cdot K^{-1} \cdot mol^{-1}$
R	8.314	$J \cdot K^{-1} \cdot mol^{-1}$

Final Summary: Solubility of Gases in Liquids

Aspect	Detail
Henry's Law Applications	Carbonated beverages, deep sea diving, aquatic life
Temperature Effect	Gas solubility **decreases** with increasing temperature
Pressure Effect	Gas solubility **increases** with increasing pressure

Environmental Application: Rising water temperatures reduce dissolved oxygen, causing anoxia in aquatic ecosystems.

Pedagogical Insights from the Instructor

Insight	Application
Smart Preparation	Focus on PYQs and predictable patterns rather than covering everything
Time Management	Lecture series designed to be concise to allow time for self-practice
Conceptual Clarity	Understanding why formulas work and when to apply which version
Common Errors	Neglecting solute moles in denominator; confusing approximate vs exact formulas; not recognizing when van't Hoff factor is needed
Exam Strategy	Easy questions must not be missed; tough questions differentiate top rankers; moderate questions are less consequential

The chapter concludes with the van't Hoff factor and abnormal molar masses, which modifies all previous colligative property formulas when solutes dissociate or associate in solution.

🎯 Putting it All Together

🔗 Connecting the Core Themes

> **Solutions** are homogeneous mixtures whose behavior is governed by the nature of the **solvent**, the **solute**, and their **intermolecular interactions**.

- **Classification** – All three sources agree that solutions are grouped by the **physical state of the solvent** (gaseous, liquid, solid), but the JEE syllabus narrows the focus to **liquid solutions** where binary mixtures dominate.
- **Solubility & "Like Dissolves Like"** – The maximum amount of solute that can dissolve (solubility) is a function of **temperature, pressure, and polarity match**. Endothermic dissolution of solids increases with temperature, whereas gas solubility follows **Henry's law** and drops as temperature rises.

📈 Vapor-Pressure Relationships

Concept	Expression	Key Insight
Vapor pressure of a pure liquid	$P = K_p$ (equilibrium constant)	Depends **only on temperature**; intensive property.
Raoult's law (non-volatile solute)	$P = P^0 \cdot X_{solvent}$	Leads to **relative lowering of vapor pressure** = X_{solute}.
Raoult's law (both volatile)	$P_{total} = P^0\_A \cdot X_A + P^0\_B \cdot X_B$	Combined with **Dalton's law** to obtain vapor-phase composition (Y).
Henry's law (gas in liquid)	$p = K_h \cdot x$	K_h rises with temperature → **solubility falls**.

- **Ideal solutions** obey Raoult's law across the entire composition range ($\Delta H_{mix} = \Delta V_{mix} = 0$).

- **Non-ideal solutions** show **positive** or **negative deviations**, reflected in altered vapor pressures and the formation of **azeotropes**.

⚖️ Ideal vs. Non-Ideal Behaviour

Deviation	Vapor-pressure trend	ΔH_{mix}	ΔV_{mix}	Typical example
Positive	$P_{obs} > P_{calc}$	**> 0** (endothermic)	**> 0** (expansion)	Ethanol + Water
Negative	$P_{obs} < P_{calc}$	**< 0** (exothermic)	**< 0** (contraction)	HCl + Water

These deviations dictate whether a mixture forms a **minimum-boiling azeotrope** (positive) or a **maximum-boiling azeotrope** (negative).

🧪 Colligative Properties – Unified Formulas

Property	Base formula	With van't Hoff factor (i)
Relative lowering of vapor pressure	$(P° - P)/P° = X_{solute}$	$(P° - P)/P° = i{\cdot}X_{solute}$
Boiling-point elevation	$\Delta T_b = K_b{\cdot}m$	$\Delta T_b = i{\cdot}K_b{\cdot}m$
Freezing-point depression	$\Delta T_f = K_f{\cdot}m$	$\Delta T_f = i{\cdot}K_f{\cdot}m$
Osmotic pressure	$\pi = C{\cdot}R{\cdot}T$	$\pi = i{\cdot}C{\cdot}R{\cdot}T$

- **K_b** and **K_f** are solvent-specific constants derived from R, the boiling/freezing temperatures, molar mass of the solvent, and the respective latent heats.
- The **van't Hoff factor** captures **dissociation** ($i = 1 + (n - 1)\alpha$) or **association** ($i = 1 - \alpha + \alpha/n$), modifying every colligative expression.

🌡️ Practical Applications Highlighted

- **Carbonated drinks** – CO_2 dissolved under high pressure; opening the bottle lowers pressure → Henry's law predicts bubble formation.
- **Scuba diving** – High ambient pressure raises N_2 solubility; rapid ascent breaches Henry's equilibrium, causing "the bends."
- **Reverse osmosis** – Applying pressure > π forces water through a semi-permeable membrane, enabling desalination.
- **Azeotropic distillation** – Minimum-boiling azeotropes (e.g., 95 % ethanol) cannot be separated by simple fractional distillation; special techniques

(extractive distillation, pressure-swing) are required.

📚 Study-Strategy Takeaways

- **Master the four colligative formulas** and the conditions under which each applies.
- **Identify the deviation type** (positive/negative) early in a problem; it decides whether you'll use Raoult's law directly or account for azeotropic behavior.
- **Practice with JEE-style questions** that combine concepts (e.g., a solute that partially dissociates *and* associates) to cement the use of the van't Hoff factor.
- **Link vapor-pressure graphs** to the underlying thermodynamics: a straight line → ideal, a curvature → deviation, a tangent point → azeotrope.

By weaving together the classification of solutions, the thermodynamic foundations of vapor pressure, the nuanced behavior of ideal vs. non-ideal mixtures, and the universal nature of colligative properties, the three sources collectively provide a **cohesive roadmap** for mastering the Solutions chapter in Physical Chemistry.